Wind Energy Fundamentals

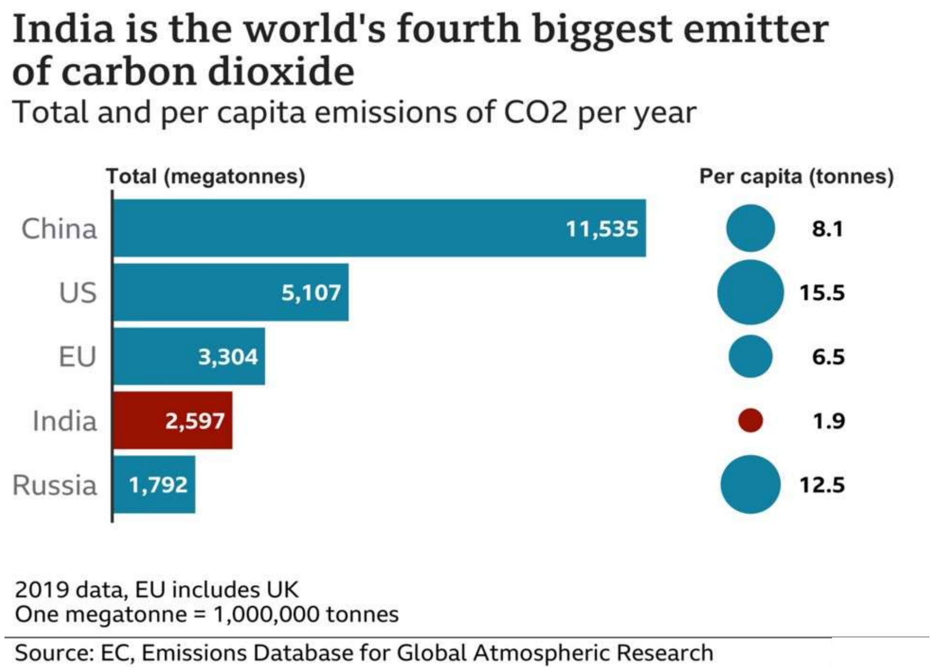
Indian Wind Energy Scenario

Wind Energy Resource and Potential

India’s Commitments at COP26 (Glasgow)

- To raise the non-fossil fuel-based energy capacity of the country to 500 GW by 2030
- To meet 50% of the country’s energy requirements using renewable energy sources by 2030
- To reduce the total projected carbon emission by one billion tonnes between now and 2030
- To reduce 45% carbon intensity by 2030, and
- To become carbon neutral and achieve net-zero emissions by 2070

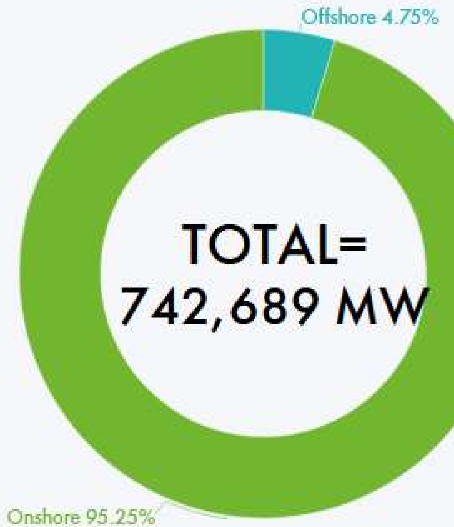
2016 (Paris agreement)
2030
40% of tot. ele
non-fossil.



450 GW
Ren. energy
2030

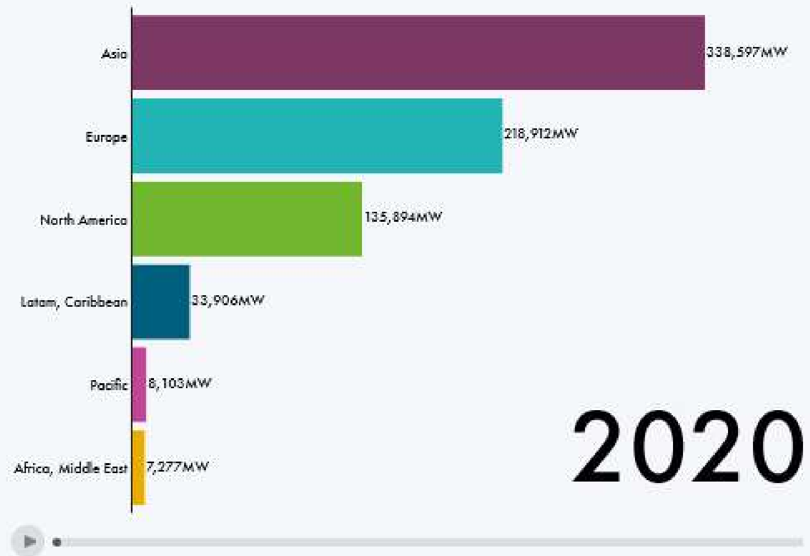
2018
CCA
57%
2027
Ren. En $\boxed{175\ \text{GW}\ 2022}$

Cumulative global wind power installations by end of 2020



GWEC
GLOBAL WIND ENERGY COUNCIL

Growth of wind power capacity from 2021-2025 by region



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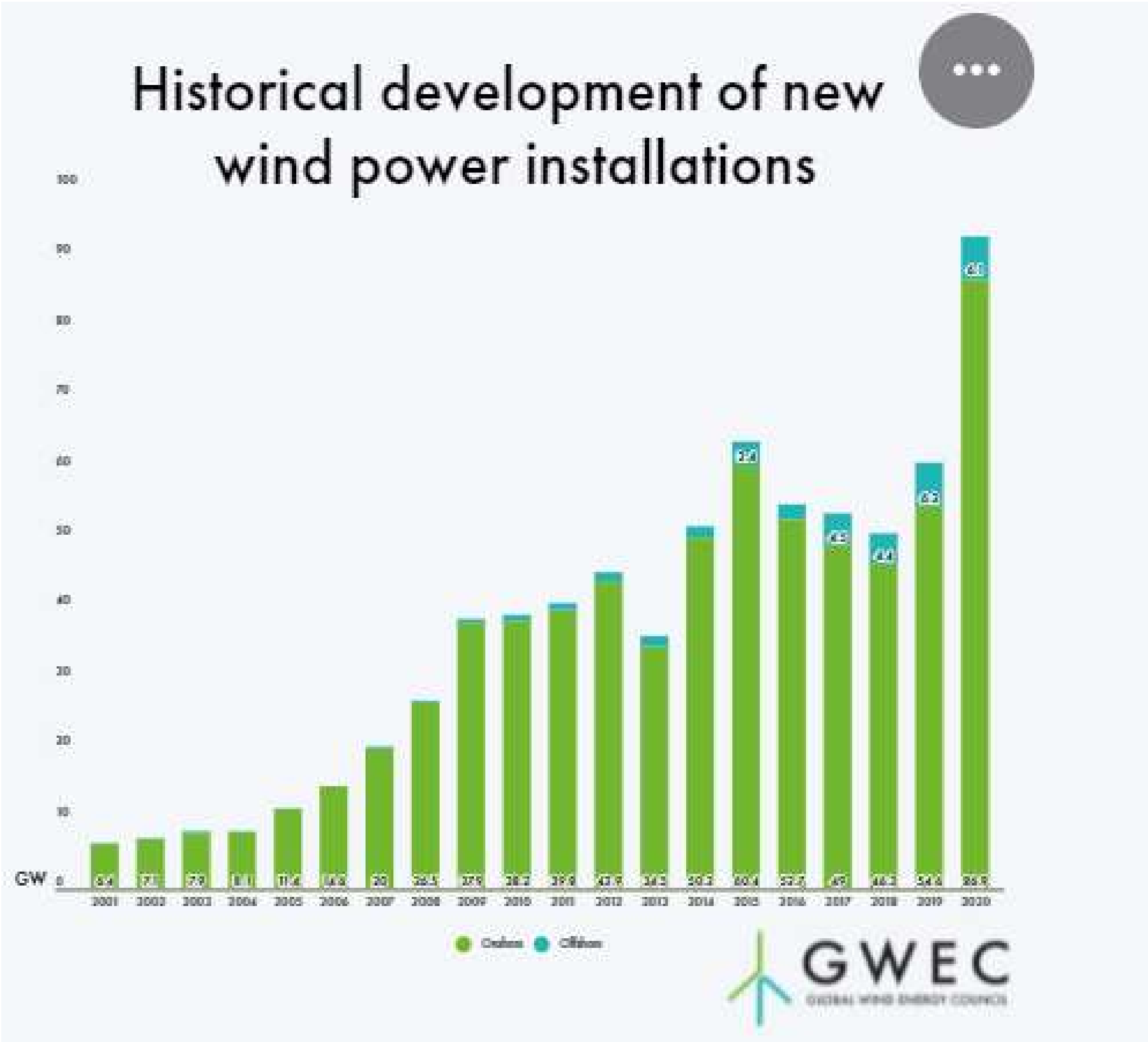
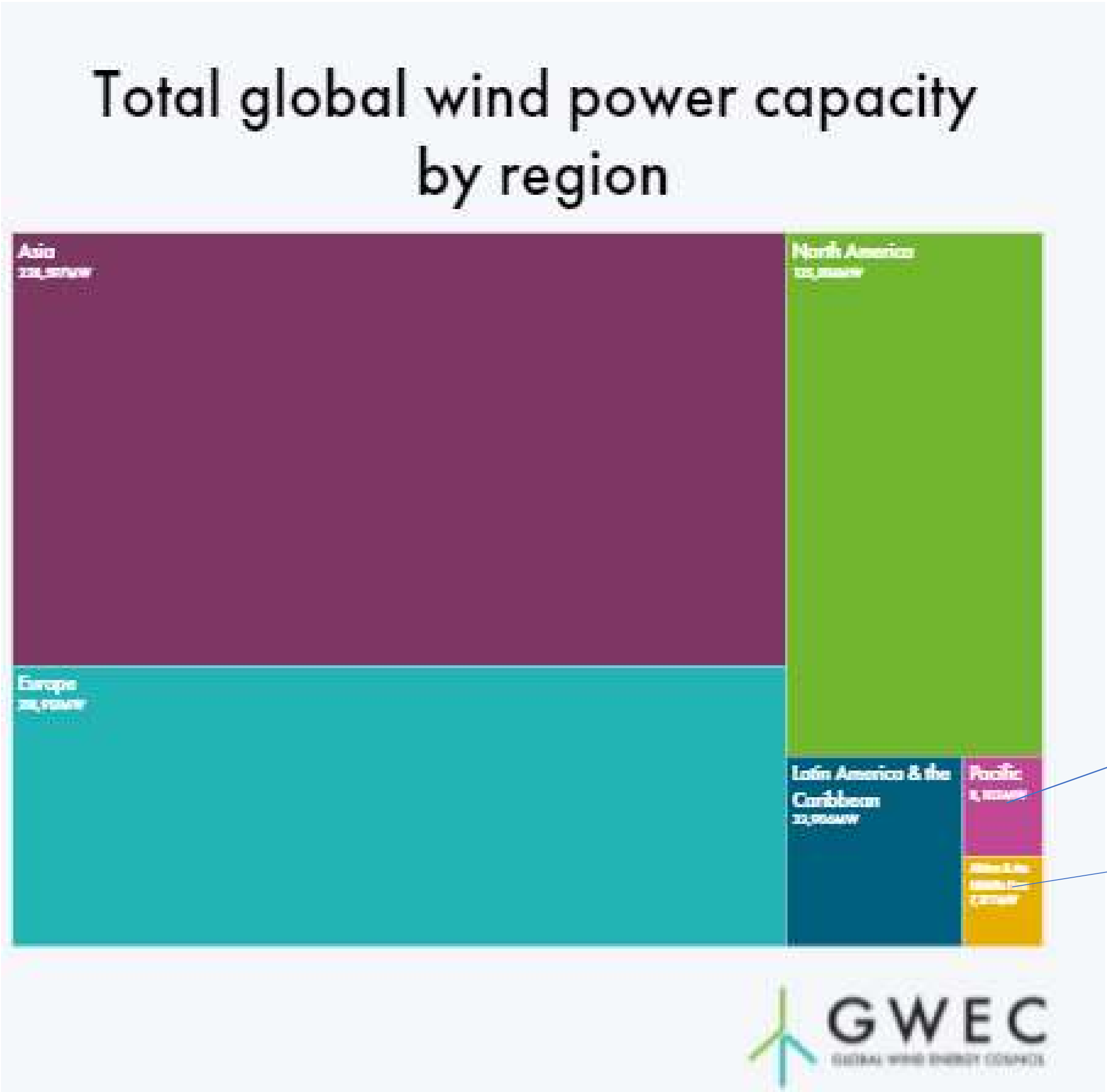

Historical development of new wind power installations
GW
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
GWEC



Pacific

Africa and Middle East

Currently has the fourth highest wind installed capacity in the world with total installed capacity of 39.25 GW (as on 31st March 2021)
Expansion of the wind industry has resulted in a strong ecosystem, project operation capabilities and manufacturing base of about 10,000 MW per annum
Generated around 60.149 Billion Units during 2020-21

Promotion by Government
Private sector investment by providing various fiscal and financial incentives such as Accelerated Depreciation benefit concessional custom duty exemption on certain components of wind electric generators
Generation Based Incentive (GBI) Scheme was available for the wind projects commissioned before 31 March 2017
Technical support including wind resource assessment and identification of potential sites through the National Institute of Wind Energy, Chennai
To facilitate inter-state sale of wind power, the inter-state transmission charges and losses have been waived off for wind and solar projects to be commissioned by March, 2022
Issued Guidelines for Tariff Based Competitive Bidding Process for Procurement of Power from Grid Connected Wind Power Projects - enable the distribution Licensees to procure wind power at competitive rates in a cost effective manner

Potential of Wind Energy in India

National Institute of Wind Energy (NIWE), has installed over 800 wind-monitoring stations all over country

NIWE issued wind potential maps at 50m, 80m, 100m and 120m above ground level

Recent assessment indicates a gross wind power potential of

302 GW in the country at 100 meter and

695.50 GW at 120 meter above ground level

S. No.	State	Wind Potential at 100 m (GW)	Wind Potential at 120 m (GW)
1	Gujarat	84.43	142.56
2	Rajasthan	18.77	127.75
3	Maharashtra	45.39	98.21
4	Tamil Nadu	33.79	68.75
5	Madhya Pradesh	10.48	15.40
6	Karnataka	55.85	124.15
7	Andhra Pradesh	44.22	74.90
	Total 7 windy states	292.97	651.72
8	Others	9.28	43.78
	Total	**302.25**	**695.50**

Wind Energy in India – Installed Capacity

https://mnre.gov.in/wind/current-status/

Off-shore Wind Energy in India

India is blessed with a **coastline of about 7600 km** surrounded by water on three sides and has good prospects of harnessing offshore wind energy

Nationaloffshore wind energy policy –Oct. 2015

MNRE will act as the nodal Ministry for development of Offshore Wind Energy in India and work in close coordination with other government entities for Development and Use of Maritime Space within the Exclusive Economic Zone (EEZ) of the country and shall be responsible for overall monitoring of offshore wind energy development in the country

National Institute of Wind Energy (NIWE), Chennai will be the nodal agency to carryout resource assessment, surveys and studies in EEZ, demarcate blocks and facilitate developers for setting up offshore wind energy farms.

Ministry set a target of 5.0 GW of offshore wind installations by 2022 and 30 GW by 2030 which has been issued to give confidence to the project developers in India market.

Absence of any obstruction in the sea offers much better quality of wind and its conversion to electrical energy.

Offshore wind turbines are much larger in size (in range of 5 to 10 MW per turbine) as against 2-3 MW of an onshore wind turbine

Cost per MW for offshore turbines are higher because of stronger structures and foundations needed in marine environment

Desirable tariffs can be achieved on account of higher efficiencies of these turbines after development of the eco system

Off-shore Wind Energy – Global Scenario

Global Scenario

Globally offshore wind is about two decades old history with the first offshore wind turbine in Denmark in 1991 which has been decommissioned in 2017

As of now the total installation is about **35.3 GW** in **18 different countries** of which important ones are-

UK (10428 MW), China (9996 MW), Germany (7689 MW), Denmark (1701 MW), The Netherlands (2611 MW), Belgium (2261 MW), Sweden (192 MW)

Capacity additions in **last four years stands at 57% of the total capacity** out of which 6358 MW commissioned in 2020, 4679 MW in 2019, 4792 MW in 2018 and 4495 MW in 2017

Identified Offshore Wind Zones for Initial Project Development

Based on the preliminary assessment from satellite data and data available from other sources eight zones each in **Gujarat** and **Tamil Nadu** have been identified as potential offshore zones for exploitation of offshore wind energy. Initial assessment by NIWE within the identified zones suggests **36 GW** of offshore wind energy potential exists off the coast **of Gujarat** only. Further, nearly **35 GW** of offshore wind energy potential exists off the **Tamil Nadu coast**

Measurement campaign deploying Light Detection and Ranging (LiDARs) at the identified zones off the coast of Gujarat and Tamil Nadu. One LiDAR was commissioned in November 2017 for Offshore Wind Resource assessment in identified zone-B off the coast of Gujarat nearly 25 km away from the port of Pipavav. **Two years data collected** from the deployed LiDAR has been analyzed and the report is published at NIWE's website

As per the report, the **annual average wind speed** at the locations is observed to be **7.52 m/s at 100m hub height.**

measures all essential incoming wind conditions, including rotor average wind speed, wind direction, shear, and turbulence, at distances of up to 200 meters from the turbine rotor. This enables the turbine control system to achieve the best possible efficiency of wind energy production

LIDAR positioned on the wind turbine blade

Oceanographic, Geophysical and Geotechnical studies

In addition to the wind data, the viability of offshore wind projects also largely depends on the condition of site in terms of oceanographic data, geophysical and geotechnical data.
Ministry has planned to carry out the required study in this regard through NIWE and provide the basic data to the stakeholders before commencement of the bidding so as to mitigate the risks. Geo-physical Survey for 365 Sq. km (Gujarat) for 1.0 GW project capacity in Gujrat has been completed. One rapid EIA has also been carried out for this site.

The first 1.0 GW offshore Wind Energy Project

The first offshore wind energy project of 1.0 GW capacity was planned in the identified zone-B off the coast of Gujarat in order to bring the economy of scale and localization of necessary ecosystem for offshore wind energy sector. In principle (stage-I) clearance as per the National offshore wind energy policy from relevant Ministries/Departments for a 1.0 GW offshore wind energy project in Zone –B off Gujarat coast has already been obtained. Expression of Interest (EoI) for the first offshore wind project of 1.0 GW capacity off Gujarat coast has been floated on 10.04.18 by NIWE. Thirty five International/Multinational and Indian Developers/OEMs have participated in the EoI and subsequent consultation process who provided their inputs for bidding process.
Ministry has constituted a committee to finalize a roadmap for development of offshore wind energy in the country along with the upcoming projects.